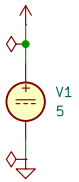
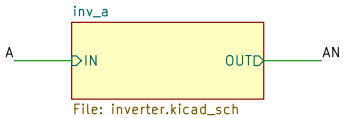
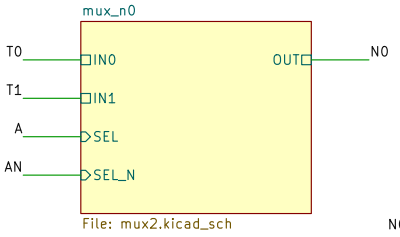
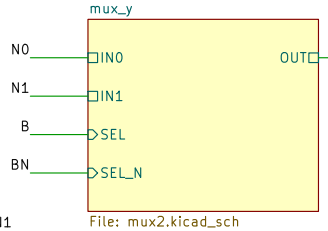
Look-up table with two inputs, built from transmission gates
Y = T[2 B + A], table T3 T2 T1 T0 = 0110

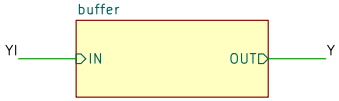
Supply



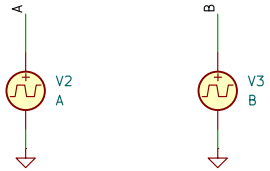



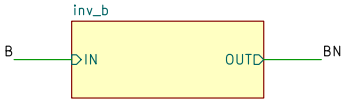



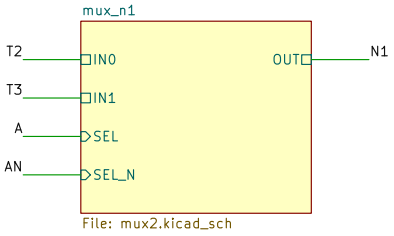




Inputs A and B




Load

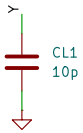
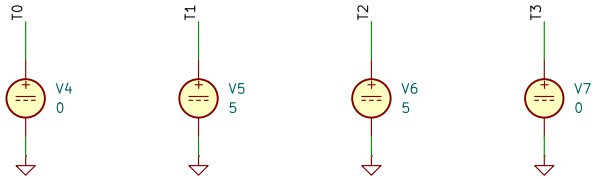
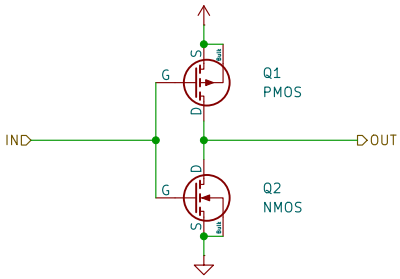


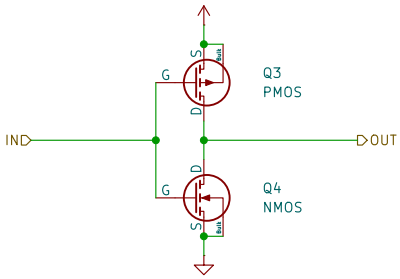

Table bits T0..T3 (in an FPGA: SRAM cells)



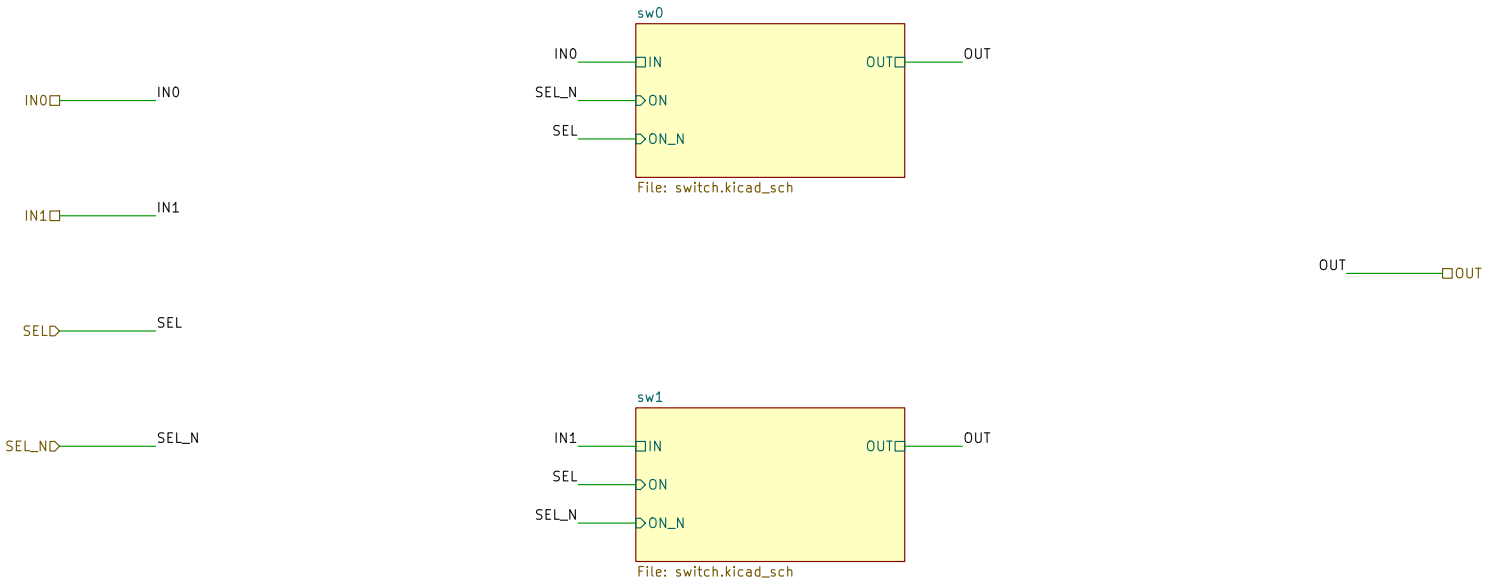
CMOS inverter: OUT = not IN
The PMOS conducts for IN = 0, the NMOS for IN = 1.



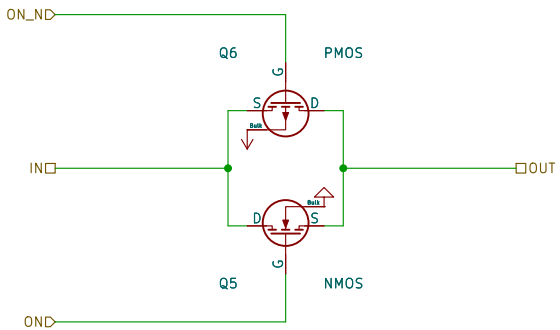
CMOS inverter: OUT = not IN
The PMOS conducts for IN = 0, the NMOS for IN = 1.



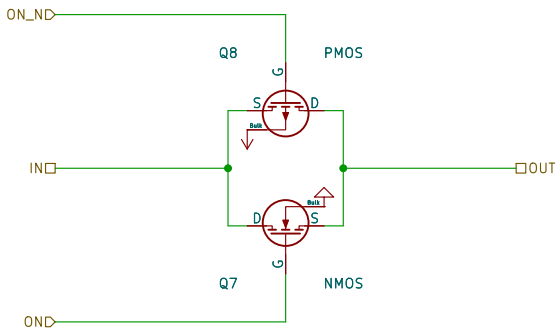
2:1 multiplexer: OUT = IN1 if SEL = 1, otherwise IN0.
Exactly one of the two switches is closed.



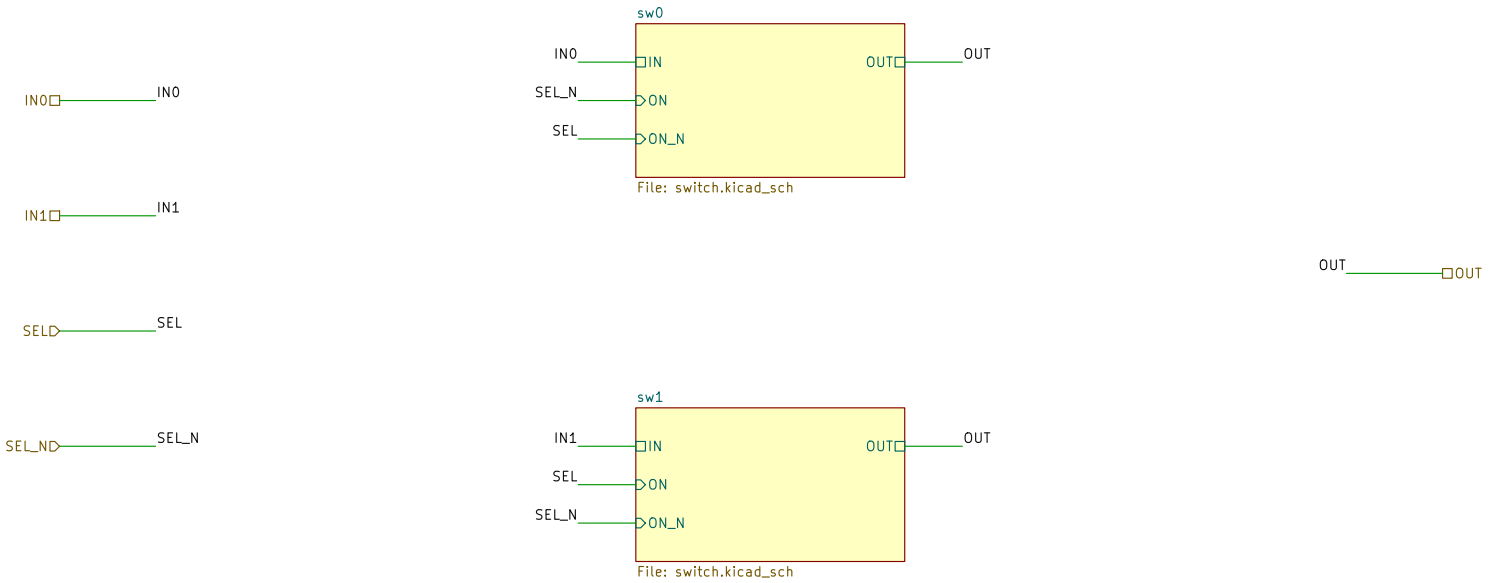
Transmission gate: IN and OUT are connected when ON = 1 and ON_N = 0.
The NMOS passes a 0 well, the PMOS passes a 1 well; together they pass both.



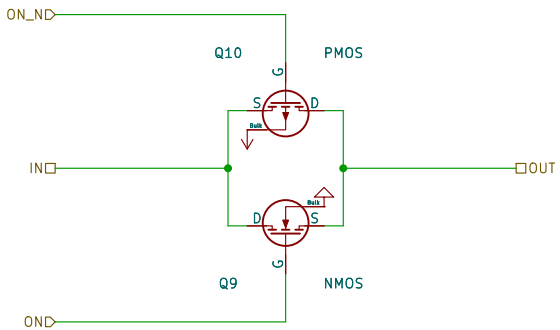
Transmission gate: IN and OUT are connected when ON = 1 and ON_N = 0.
The NMOS passes a 0 well, the PMOS passes a 1 well; together they pass both.



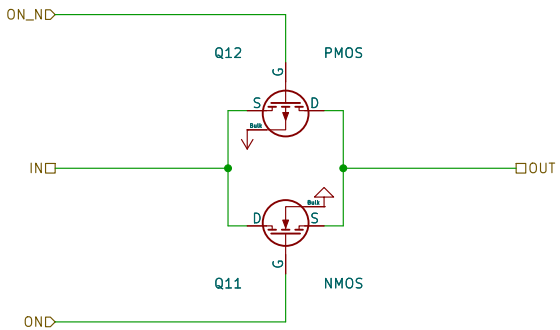
2:1 multiplexer: OUT = IN1 if SEL = 1, otherwise IN0.
Exactly one of the two switches is closed.



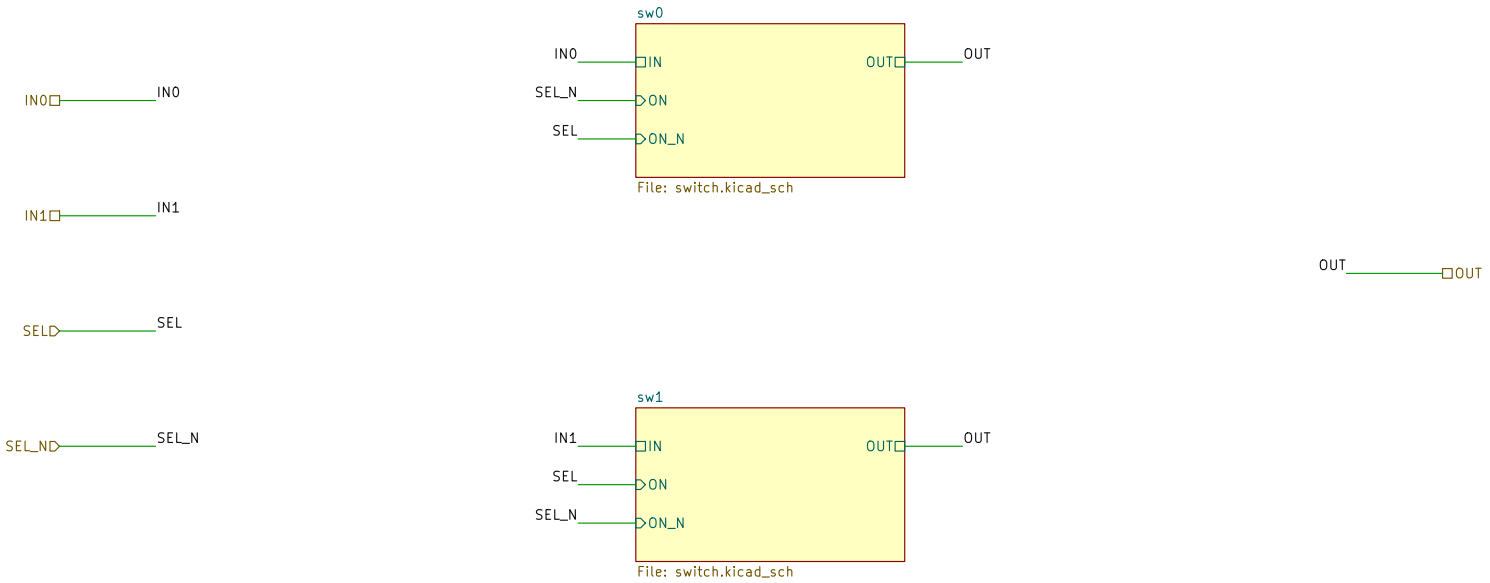
Transmission gate: IN and OUT are connected when ON = 1 and ON_N = 0.
The NMOS passes a 0 well, the PMOS passes a 1 well; together they pass both.



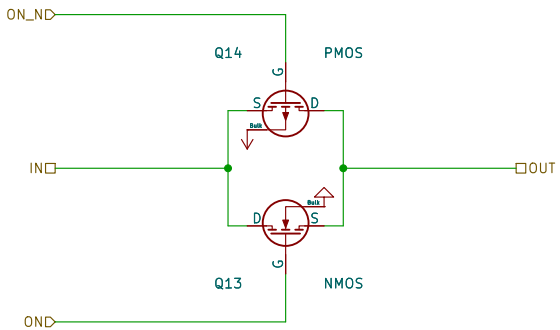
Transmission gate: IN and OUT are connected when ON = 1 and ON_N = 0.
The NMOS passes a 0 well, the PMOS passes a 1 well; together they pass both.



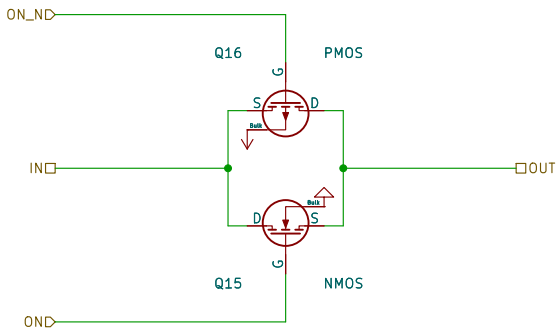
2:1 multiplexer: OUT = IN1 if SEL = 1, otherwise IN0.
Exactly one of the two switches is closed.



Transmission gate: IN and OUT are connected when ON = 1 and ON_N = 0.
The NMOS passes a 0 well, the PMOS passes a 1 well; together they pass both.



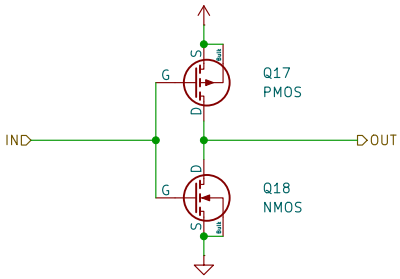
Transmission gate: IN and OUT are connected when ON = 1 and ON_N = 0.
The NMOS passes a 0 well, the PMOS passes a 1 well; together they pass both.



Output buffer: two inverters in series.
The first one restores the logic level, the second one restores the polarity.



CMOS inverter: OUT = not IN
The PMOS conducts for IN = 0, the NMOS for IN = 1.



CMOS inverter: OUT = not IN
The PMOS conducts for IN = 0, the NMOS for IN = 1.

